Lecture 3: Population Inference for Means

August 24, 2023

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- We saw that for large enough sample sizes, our sample proportion was likely to be close to the population proportion.
- We also demonstrated that we could predict how close we might by using confidence intervals.
- We explored a bootstrapping approach to producing confidence intervals.

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- In this lecture, we consider variables that are not binary and our goal is to learn about the population mean (not just a proportion) from a random sample (with replacement) from the population.
- Because the variables we will consider may take on many values, we will not know the type of probability distribution underlying our data (e.g., the Bernoulli distribution in the binary variable case).
- Not knowing the distribution isn't a problem when the sample is large enough, but understanding why it isn't a problem is subtle.

The Current Population Survey

Something about the CPS sampling here...
Recall that our sample mean based on 2,258 working age Californians is $135,406

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Two Questions

Unlike the turnout example, we will never know the true population income mean, which we will denote with μ. We will also never know the population income standard deviation, which we will denote σ or the population variance which we denote with σ^2. We also will not be able to generate a sampling distribution. But we still want to answer the two questions.

1. Is \$135,406 likely to be close to the population mean μ?
2. What can we say about how close?

Sampling Distribution for the Sample Mean

i	1	2	...	2258	
s	Y_1	Y_2	...	Y_{2258}	$\overline{Y}_{2258}$
1	37,006	108,811	...	49,001	135,406
$f.(\cdot)$	?	?	...	?	?
$E.(\cdot)$	μ	μ	...	μ	?
$V.(\cdot)$	σ^2	σ^2	...	σ^2	?

- Recall that we can think of $Y_1, ..., Y_n$ as *i.i.d.* RVs.
- Although we don't know the distribution $f_Y(y)$ of these RVs, their expected value $E[Y]$ is the population mean μ and their variance $V[Y]$ is the population variance σ^2.

Sampling Distribution for the Sample Mean

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s	Y_1	Y_2	...	Y_{2258}	$\overline{Y}_{2258}$
1	37,006	108,811	...	49,001	135,406
2	?	?	...	?	?
3	?	?	...	?	?
4	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	?	?	...	?	?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f_{\cdot}(\cdot)$	?	?	...	?	?
$E_{\cdot}(\cdot)$	μ	μ	...	μ	?
$V_{\cdot}(\cdot)$	σ^2	σ^2	...	σ^2	?

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What can we learn?

i	1	2	$\dots$	2258	
s	Y_1	Y_2	$\dots$	Y_{2258}	$\overline{Y}_{2258}$
1	37,006	108,811	$\dots$	49,001	135,406
2	?	?	$\dots$	?	?
3	?	?	$\dots$	?	?
4	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	?	?	$\dots$	?	?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f_\cdot(\cdot)$	?	?	$\dots$	?	?
$E_\cdot(\cdot)$	μ	μ	$\dots$	μ	?
$V_\cdot(\cdot)$	σ^2	σ^2	$\dots$	σ^2	?

- Although we can't observe the population or the sampling distribution, we can mathematically derive certain facts about the sampling distribution.
- We will focus on the three red question marks.

The Type of Distribution

i	1	2	...	2258	
s	Y_1	Y_2	...	Y_{2258}	$\overline{Y}_{2258}$
1	37,006	108,811	...	49,001	135,406
2	?	?	...	?	?
3	?	?	...	?	?
4	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	?	?	...	?	?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f_\cdot(\cdot)$	?	?	...	?	N(?,?)
$E_\cdot(\cdot)$	μ	μ	...	μ	?
$V_\cdot(\cdot)$	σ^2	σ^2	...	σ^2	?

Property #1: *The sampling distribution of the sample mean is approximately normal when n is large.*

- The approximate normality result comes from the central limit theorem (CLT).
- We know that $\overline{Y}_{2258} \sim^{approx} N(?, ?)$, but still don't know the mean or the variance of this distribution.

What is the $E[\overline{Y}_n]$?

i	1	2	$\dots$	2258	
s	Y_1	Y_2	$\dots$	Y_{2258}	$\overline{Y}_{2258}$
1	37,006	108,811	$\dots$	49,001	135,406
2	?	?	$\dots$	?	?
3	?	?	$\dots$	?	?
4	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	?	?	$\dots$	?	?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f_\cdot(\cdot)$	?	?	$\dots$	?	$N(?, ?)$
$E_\cdot(\cdot)$	μ	μ	$\dots$	μ	?
$V_\cdot(\cdot)$	σ^2	σ^2	$\dots$	σ^2	?

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1	37,006	108,811	...	49,001	135,406
2	?	?	...	?	?
3	?	?	...	?	?
4	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	?	?	...	?	?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f_{\cdot}(\cdot)$	?	?	...	?	$N(?, ?)$
$E_{\cdot}(\cdot)$	μ	μ	...	μ	?
$V_{\cdot}(\cdot)$	σ^2	σ^2	...	σ^2	?

Property #1: *The sample mean is unbiased for the population mean.* The expected value of the sample mean, $\mathrm{E}[\overline{Y}] = \mu$.

Proof:

$$\begin{aligned}
\mathrm{E}[\overline{Y}] &= \mathrm{E}\left[\frac{1}{n}(Y_1 + ... + Y_n)\right] \\
&= \\
&= \\
&= \\
&= \\
&= \mu.
\end{aligned}$$

Sample mean is unbiased for the population mean

i	1	2	$\dots$	2258	
s	Y_1	Y_2	$\dots$	Y_{2258}	$\overline{Y}_{2258}$
1	37,006	108,811	$\dots$	49,001	135,406
2	?	?	$\dots$	?	?
3	?	?	$\dots$	?	?
4	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	$\dots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	?	?	$\dots$	?	?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f_.(\cdot)$	?	?	$\dots$	?	$N(\mu, ?)$
$E_.(\cdot)$	μ	μ	$\dots$	μ	μ
$V_.(\cdot)$	σ^2	σ^2	$\dots$	σ^2	?

What is the $V[\overline{Y}_n]$?

i	1	2	$\ldots$	2258	
s	Y_1	Y_2	$\ldots$	Y_{2258}	$\overline{Y}_{2258}$
1	37,006	108,811	$\ldots$	49,001	135,406
2	?	?	$\ldots$	?	?
3	?	?	$\ldots$	?	?
4	$\vdots$	$\vdots$	$\ldots$	$\vdots$	$\vdots$
5	$\vdots$	$\vdots$	$\ldots$	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	?	?	$\ldots$	?	?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f.(\cdot)$	?	?	$\ldots$	?	N(μ,?)
$E.(\cdot)$	μ	μ	$\ldots$	μ	μ
$V.(\cdot)$	σ^2	σ^2	$\ldots$	σ^2	?

Property #2: The *sampling variance* of the sample mean,

$$V[\overline{Y}] = \frac{\sigma^2}{n}$$

Proof:

$$\begin{aligned}
V[\overline{Y}] &= V\left[\left(\frac{1}{n}Y_1 + ... + \frac{1}{n}Y_n\right)\right] \\
&= \\
&= \\
&= \\
&= \\
&= \frac{\sigma^2}{n}.
\end{aligned}$$

How much does $\overline{Y}_{2258}$ vary from sample to sample?

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1	37,006	108,811	...	49,001	135,406
2	?	?	...	?	?
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5	$\vdots$	$\vdots$	...	$\vdots$	$\vdots$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
1 mil	?	?	...	?	?
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
$f.(\cdot)$	?	?	...	?	$N(\mu, \frac{\sigma^2}{n})$
$E.(\cdot)$	μ	μ	...	μ	μ
$V.(\cdot)$	σ^2	σ^2	...	σ^2	$\frac{\sigma^2}{n}$

Facts about the sampling distribution of $\overline{Y}_n$

What we know about the sampling distribution of $\overline{Y}_n$:

1. Approximately normally distributed
2. Centered at μ.
3. Variance of $\frac{\sigma^2}{n}$

These facts imply that the middle 95% of the distribution lies approximately between $\mu - 1.96 \cdot \frac{\sigma}{\sqrt{n}}$ and $\mu + 1.96 \cdot \frac{\sigma}{\sqrt{n}}$. As long as $\sqrt{n}$ is big in relation to σ, $\overline{Y}_n$ will likely be close to μ.

We still don't know:

1. The value of μ.
2. The value of σ.

Two Questions

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Two Questions

1. Is \$135,406 close to the population mean μ? Yes!
2. Can say something about how close? Let's use a confidence interval. We will look at two methods: bootstrapping and plug-in.

Bootstrapped Sampling Distribution of a Sample Mean

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y_i	37,006	108,811	...	49,001	135,406

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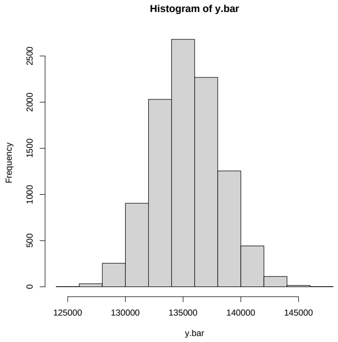
With the data stored in the vector y, each mean of the bootstrapped sampling distribution can be generated with

```
mean(sample(y, size=2258, replace=TRUE))
```

Bootstrapped CIs for a Population Mean

```
B=10000
y.bar = rep(NA,B)
for(b in 1:B){y.bar[b]= mean(sample(y,size=n,replace=T))}
hist(y.bar)
```




```
> quantile(y.bar,probs=c(.025,.975)) ## bootstrapped CI
    2.5%    97.5%
129783.2 141169.0
```

Bootstrapped Sampling Distribution vs Sampling Distribution

1. CLT says the sampling distribution is approximately normal.
2. We see that the bootstrapped distribution is approximately normal.
3. Remaining question is whether the bootstrapped CI has approximately the same size as the middle 95% of the sampling distribution.

Let's table the remaining question for now while we look at another approach to CIs.

Plug-in CI

- Another way to get a 95% CI is to use the fact that the middle 95% of the distribution lies approximately between $\mu - 1.96 \cdot \frac{\sigma}{\sqrt{n}}$ and $\mu + 1.96 \cdot \frac{\sigma}{\sqrt{n}}$.
- If we denote the sample standard deviation with s, then we plug in sample values into the above statement: $\overline{y} - 1.96 \cdot \frac{s}{\sqrt{n}}$ and $\overline{y} + 1.96 \cdot \frac{s}{\sqrt{n}}$
- 129,720 and 141,091 (note how close these are to the bootsrapped CI)

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Why does this work?

The facts we've derived about the sampling distribution of the sample mean tell us that

$$P(\mu - 1.96 \cdot \frac{\sigma}{\sqrt{n}} \leq \overline{Y} \leq \mu + 1.96 \cdot \frac{\sigma}{\sqrt{n}}) \approx .95$$

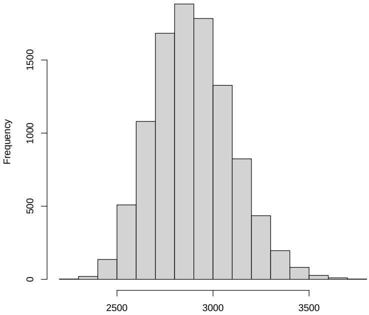
Some algebra (left as an exercise) shows that this implies

$$P(\overline{Y} - 1.96 \cdot \frac{\sigma}{\sqrt{n}} \leq \mu \leq \overline{Y} + 1.96 \cdot \frac{\sigma}{\sqrt{n}}) \approx .95$$

Remaining question is whether $\frac{\sigma}{\sqrt{n}} \approx \frac{s}{\sqrt{n}}$ (note how similar this is to the remaining bootstrapped CI question).

Partially Addressing Remaining Question

- $\frac{\sigma}{\sqrt{n}}$ is called a standard error because it is the standard deviation for an estimator ($\overline{Y}$) of a parameter (μ).
- $\frac{s}{\sqrt{n}}$ is called an estimated standard error. We will bootstrap this estimated standard error to get a sense of its uncertainty (note the scale of this uncertainty).



CI Wrap Up

We got the following 95% CIs for the population income mean:

- Boostrap: $129,783 to $141,169
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- Plug-in: $129,720 to $141,091

From the previous slide, we see the estimated standard error is very unlikely to be off by more than $1,000, and therefore the bounds are very unlikely to be off by more than $2,000.

We are at least 95% confident that the true population mean is roughly between $128,000 and $143,000, despite observing only a tiny fraction of the population.

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1. Is \$135,406 close to the population mean μ? Yes!
2. Can say something about how close? Yes! Likely within \$7,500 or so.

Summary

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